

Advanced (Habanero)

- Q1.** Solve the system: $x + y = 3$, $x^2 + y^2 = 9$
 (A) $(x, y) = (1, 2)$
 (B) $(x, y) = (2, 1)$
 (C) $x = 1$, $y = 2$ or $x = 2$, $y = 1$
 (D) $x = -1$, $y = 4$ or $x = 4$, $y = -1$
 (E) No solution
- Q2.** Find the intersection point of $y = x^2 - 2x + 1$ and $y = -x + 3$
 (A) $(1, 2)$
 (B) $(2, 1)$
 (C) $(1, 3)$
 (D) $(3, 1)$
 (E) $(0, 3)$
- Q3.** Solve the system: $y = x^2 + 2x - 3$, $y = x - 1$
 (A) $x = -3$, $y = -4$
 (B) $x = -1$, $y = -2$
 (C) $x = 1$, $y = 0$
 (D) $x = 3$, $y = 4$
 (E) $x = -1$, $y = -2$ or $x = 1$, $y = 0$
- Q4.** What is the number of solutions to the system $x^2 + y^2 = 4$, $x + y = 0$?
 (A) 0
 (B) 1
 (C) 2
 (D) 3
 (E) 4
- Q5.** Solve the system: $y = x^2 - 4x + 3$, $y = x - 3$
 (A) $x = 3$, $y = 0$
 (B) $x = 1$, $y = -2$
 (C) $x = 1$, $y = -2$ or $x = 3$, $y = 0$
 (D) $x = -1$, $y = -4$
 (E) $x = -3$, $y = -6$
- Q6.** Find the point of intersection of $y = x^2 - 2$ and $y = x + 2$
 (A) $(-2, 2)$
 (B) $(2, 4)$
 (C) $(-1, 1)$
 (D) $(1, 3)$
 (E) $(-2, 2)$ or $(1, 3)$
- Q7.** Solve the system: $x^2 + y^2 = 1$, $x + y = 1$
 (A) $x = y = -\frac{1}{2}$
 (A) $x = -\frac{1}{2}$, $y = -\frac{1}{2}$
 (A) $x = -\frac{1}{2}$, $y = -\frac{1}{2}$ or $x = \frac{1}{2}$, $y = \frac{1}{2}$
 (A) $x = \frac{1}{2}$, $y = \frac{1}{2}$
 (A) $x = \frac{1}{2}$, $y = \frac{1}{2}$ or $x = -\frac{1}{2}$, $y = -\frac{1}{2}$
 (A) $x = -\frac{1}{2}$, $y = -\frac{1}{2}$
- Q8.** Find the number of solutions to the system $y = x^2 + 1$, $y = -x^2 + 1$
 (A) 0
 (B) 1
 (C) 2
 (D) 3
 (E) 4

Open Ended Questions (FRQ)

- Q9.** Solve the system: $x^2 + y^2 = 25$, $x + y = 5$. Find the intersection points and interpret the results.

- Q10.** Consider the system of equations $y = x^2 - 2x + 1$, $y = x - 1$. Find the point(s) of intersection. Write a short explanation of your method.

Chef's Tips

- Tip 1: Encourage students to graph the equations on a coordinate plane.
- Tip 2: Use the method of substitution or elimination to solve the system of equations.
- Tip 3: Emphasize the importance of checking the solutions in the original equations.